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Boats & Streams

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BOATS & STREAMS

A. Definitions

If a boat is moving along with the direction of the stream, then it is said to go downstream. The net speed of boat in this case is called downstream speed.

A boat is moving against the direction of the stream then it is said to go upstream. The net speed of boat in this case is called upstream speed.

Let the speed of the boat in still water is 'a' km/hr and the speed of the stream is 'b' km/hr. If the boat goes downstream, then the speed will be $(a + b)$ km/hr as in this case, the water will take the boat along with it.

If the boat goes upstream, then the speed will be $(a - b)$ km/hr as in this case, the water will offer resistance to the boat.

Let the downstream speed = $D = a + b$ (i)

Then the upstream speed = $U = a - b$ (ii)

Adding the two equations, we get $2a = D + U$.

$\Rightarrow a = (D + U) / 2$ which gives the speed of the boat in terms of downstream and upstream speed.

Subtracting the equation (i) and (ii), we get $D - U = 2b$

$\Rightarrow b = \frac{D - U}{2}$ which gives the speed of the stream



B. Solved Examples

1. A man can row 15 km/hr in still water. If the speed of current is 5 km/hr and it takes 30 hours to a man to row a place and back, then how far is the place?

Sol. The speed of downstream = $15 + 5 = 20$ km/hr and the speed of upstream = $15 - 5 = 10$ km/hr

Let the distance is 'x' km. We have $\frac{x}{20} + \frac{x}{10} = 30$.

Solving this equation, we get $x = 200$ km

2. A boat takes 1 hour less to travel 120 km downstream than to travel 96 km upstream. If speed of the current is 4 km/hr, find the speed of the boat in still water.

Sol. Let the speed of boat in still water be 'x' km/hr.

We have $\frac{96}{x-4} - \frac{120}{x+4} = 1$

$$\Rightarrow 96(x+4) - 120(x-4) = x^2 - 16$$

$$\Rightarrow 96x + 384 - 120x + 480 = x^2 - 16$$

$$\Rightarrow 864 - 24x = x^2 - 16 \Rightarrow x^2 + 24x - 880 = 0$$

$$\Rightarrow x^2 + 44x - 20x - 880 = 0$$

$$\Rightarrow x(x+44) - 20(x+44) = 0$$

$$\Rightarrow (x+44)(x-20) = 0 \Rightarrow x = 20 \text{ km/hr}$$

(Rejecting the negative value)



3. A boat travels at a speed of 30 km/hr with the stream and 18 km/hr against the stream. What is the boat's speed in still water and the speed of the stream?

Sol. Speed of boat in still water = $\frac{30+18}{2} = 24\text{kmph}$ and Speed of the stream = $\frac{30-18}{2} = 6\text{kmph}$

4. If Anshul rows 40 km upstream and 120 km downstream taking 4 hours each time, then what is the speed of the stream?

Sol. Basic Formula:

Speed of the stream = $\frac{1}{2} (D - U)$ where D = speed downstream, U = speed upstream

$$\text{Speed downstream} = \frac{\text{distance travelled}}{\text{time taken}} = \frac{120}{4}$$

$$\Rightarrow D = 30\text{km / hr}$$

$$\text{Speed upstream} = 40/4 = 10 \Rightarrow U = 10 \text{ km/hr}$$

Speed of the stream

$$= \frac{1}{2} (D - U) \text{ km/hr} = \frac{1}{2} (30 - 10) = 10 \text{ km/hr}$$

5. A man rows 750 m in 675 seconds against the stream and returns in $7\frac{1}{2}$ minutes. What is the speed of boat in still water?

Sol. Speed in still water = $\frac{1}{2} (D + U)$ km/h where 'D' is speed downstream and 'U' is speed upstream.



$$\text{Speed upstream 'U'} = \frac{750}{675} = \frac{10}{9} \text{ m/sec}$$

$$\text{Speed downstream 'D'} = \frac{750}{450} = \frac{5}{3} \text{ m/s}$$

Speed of boat in still water =

$$\frac{1}{2}(D+U) = \frac{1}{2}\left(\frac{10}{9} + \frac{5}{3}\right) \times \frac{18}{5} = 5 \text{ km/hr}$$

6. A motorboat travelled a distance of 18 km downstream in 3 hours and returns back to the same place upstream in 6 hrs. What is the speed of the boat in still water?

Sol. Rate downstream = $18/3 = 6$ km/hr, Rate upstream = $18/6 = 3$ km/hr

$$\text{Speed in still water} = 1/2 \times (6 + 3) = 4.5 \text{ km/hr}$$

7. The speed of a motorboat in still water is 20 km/hr and the rate of current is 5 km/hr. The distance travelled upstream in 10 minutes is:

Sol. Speed upstream = $(20 - 5)$ km/hr = 15 km/hr.

$$\text{Distance travelled in 10 minutes} = 15 \times 10/60 = 2.5 \text{ km}$$

8. A man goes 4 km against the river in 1 hour and goes 2 km along the river in 15 minutes. How long will it take to cover 15 km in still water?

Sol. Rate upstream = 4 km/hr

$$\text{Rate downstream} = 2/(15/60) = 8 \text{ km/hr}$$

$$\text{Speed in still water} = 1/2 \times (4 + 8) = 6 \text{ km/hr}$$

$$\text{So, required time} = 15/6 = 2.5 \text{ hrs} = 2 \text{ hrs } 30 \text{ min}$$

RACES

A. Definitions

A race is a competition in which the competitors have to cover a given distance in the minimum possible time. The ground or path on which a race is run is called a racecourse. The point from where a race starts is called the starting point, and the point where race finishes is the winning point or goal.

As Races and Games is a sub-topic of Time, Speed and Distance, so all the formulas of Time, Speed and Distance are applicable to questions of Races and Games.

B. Some terminologies used in Races

P gives Q a start of x meters: This statement implies that, while P starts the race from starting point, whereas, Q starts x meters ahead of P. To cover a race of 100 meters in this case, P will have to cover 100 meters while Q will have to cover only $(100 - x)$ meters.

P beats Q by x meters: This statement implies that in the same time, while P reached the winning point, whereas, Q is behind P by x m. To cover a race of 100 meters in this case, P has covered 100 meters while Q has covered only $(100 - x)$ meters.

P can give Q a start of t minutes: This statement implies that P will start t minutes after Q starts from the starting point. Both P and Q will reach the finishing point at the same time.

P can give Q x meters or t seconds: This statement implies that P can give Q either x meters or t seconds from where you can find the speed of Q as x/t meters per second.



Dead Heat: A dead heat situation arises when all participants reach the finishing point at the same time.

C. Circular Tracks

Circular tracks may involve two or three participants. They usually start from the same point and run in the same or opposite directions. Let the length of the track be 'd' metres and if radius of circle is 'r'(given) then length of the track will be $2\pi r$.

(i) When two persons start from the same point and run in opposite directions with speeds 'x' and 'y' m/s, the following rules apply:

Their relative speed is always equal to sum of their individual speeds.

Between any two meetings, the total distance covered by them together **is always** equal to the length of the circular track.

Time taken for the first meeting after starting = $\frac{d}{x+y}$

Time taken for them to meet at the starting point for the first

time = LCM of $\left[\left(\frac{d}{x}\right) \text{ and } \left(\frac{d}{y}\right)\right]$

(ii) When two persons start from the same point and run in the same direction with speeds 'x' and 'y' m/s (assume $x > y$), the following rules apply:

Their relative speed is always equal to the difference of their individual speeds.



Between any two meetings, the faster contestant covers an extra distance equal to the length of the circular track compared to the slower contestant.

Time taken for the first meeting after starting = $d/(x - y)$

Time taken for them to meet at the starting point for the first time = LCM of $[(d/x) \text{ and } (d/y)]$.

D. Solved Examples

1. In a 400 meter race, A can give B a start of 30 meters and C a start of 67 meters. What start can B give C?

Sol. When A runs 400 meters, B runs 370 meters.

When A runs 400 meters, C runs 333 meters. i.e. when B runs 370 meters, C runs 333 meters.

Therefore, when B runs 400 meters, C runs $\frac{333 \times 400}{370} = 360$ meters.

Or B can give C a start of 40 meters in a 400 meters race.

2. A gives B a start of 20 meter in a 200-meter race and B gives C a start of 27 meter in a 300 meter race. How much start can A give C in a km race?

Sol. **Race 1:** Length 200 meter.

When A reaches 200 meter, B reaches 180 meter.

⇒ When A reaches 100 meter B reaches 90 meter.

Race 2: Length 300 meter.

When B reaches 300 meter, C reaches 273 meter



⇒ When B reaches 100 meter, C reaches 91 meter.

Therefore, in a 100 meter race, when A finishes 100 meter

C finishes $\frac{90 \times 91}{100} = 81.9$ meter.

In other words, A can give C a start of 18.1 meter in a 100-meter race.

Therefore, in a kilometre race, A can give C a start of 181 meters.

3. A runs 5 times as fast as B. If A gives B a start of 100 m, how far must the destination be so that A and B reach it at the same time?

Sol. Ratio of the speeds of A and B = 5:1.

In a race of 5 m, A gains 4 m over B. 100 meters will be gained by A in race of $(5/4 \times 100 = 125 \text{ m})$. Thus, destination must be at a distance of 125 m from the starting point.

4. In a 10 km race, A, B, and C, each running at uniform speed, get the gold, silver, and bronze medals, respectively. If A beats B by 1 km and B beats C by 1 km, then by how many metres does A beat C?

Sol. A beats B by 1 km, means A travels 10 km in the same time that B travels 9 km.

The ratio of speeds of A and B is 10 : 9. Similarly, the ratio of speeds of B and C is 10 : 9.

⇒ The ratio of speeds of A : B : C = 100 : 90 : 81



In the same time that A travels 100 m, C travels 81 metres
⇒ In the same time that A travels 10000 m, C would travel 8100 m or A would beat C by 1900 m.

5. There is a track with a length of 100 meters and 2 people, A & B, are running around it at 25 m/min and 10 m/min respectively in the same direction. When will A and B meet at the starting point for the first time?

Sol. The time of their meeting again at the starting point will be the LCM of $100/25$ & $100/10$ i.e. 4 & 10 LCM of 4 and 10 = 20. So, after 20 minutes these people will be together at the starting point.

6. There is a track with a length of 200 meters and 2 people, A & B, are running around it at 15 m/min and 25 m/min respectively in the same direction. When will B overtake A for the 1st time?

Sol. Time = $d/(b - a)$.

Time of meeting = $200/(25 - 15) = 200/10 = 20$ min.

7. Two robots M and N running around a circular track of length 2520 m in opposite directions with initial speeds of 100 m/s and 20 m/s respectively. Whenever they meet, M's speed halves and N's speed doubles. After what time from the start will they meet for the third time?

Sol. Time taken to meet for the 1st time = $\frac{2520}{100 + 20} = 21$ sec.

Now, M's speed = 50 m/s and N's speed = 40 m/s.



$$\text{Time taken to meet for the 2}^{\text{nd}} \text{ time} = \frac{2520}{50 + 40} = 28 \text{ sec}$$

Now, M's speed = 25 m/s and N's speed = 80 m/s

$$\text{Time taken to meet for the 3}^{\text{rd}} \text{ time} = \frac{2520}{25 + 80} = 24 \text{ sec}$$

Therefore, total time = $(21 + 28 + 24) = 73 \text{ sec}$

Ex 8. A and B start jogging in a circular park of circumference 150 metre from the same point simultaneously in the same direction. A jogs at a speed of 36 km/hr and B jogs at a speed of 90 km/hr. When will they meet first time again and when will they meet at the starting point?

Sol. Circumference of the park = 150 metre.

$$\text{Speed of A} = 36 \text{ km/hr} = 36 \times \frac{5}{18} = 10 \text{ m/sec,}$$

$$\text{Speed of B} = 90 \text{ km/hr} = 90 \times \frac{5}{18} = 25 \text{ m/sec}$$

They will meet again first time when B has covered one full circle more than A. i.e. when their relative speeds completed one full circle. Their relative speed = $25 - 10 = 15 \text{ m/sec}$.

$$\text{Time taken by them to meet first time} = \frac{150}{15} = 10 \text{ sec.}$$

$$\text{Time taken by A to complete a round} = \frac{150}{10} = 15 \text{ sec.}$$

$$\text{Time taken by B to complete a round} = \frac{150}{25} = 6 \text{ sec.}$$

Time taken by them to meet first time at the starting point = LCM of $(15, 6) = 30 \text{ sec}$